Magnus Grini

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Education

M.Sc. in Artificial Intelligence | NTNU & UC Berkeley (exchange) 2025–2027

- **Thesis:** bio-inspired artificial intelligence for portfolio optimization under uncertainty.
- **Relevant coursework:** Introduction to Machine Learning, Bio-Inspired Artificial Intelligence, Artificial Intelligence Methods (reasoning and decision-making under uncertainty), Introduction to Artificial Intelligence (search and constraint optimization).

B.Sc. in Computer Science | NTNU 2022–2025

- **Thesis:** real-time motion capture and calibration for adaptive Parkinson's disease gameplay.
- **Relevant coursework:** Statistics (probability, inference, regression), Mathematical Methods 1–3 (linear algebra, real analysis, numerical methods), Applied Machine Learning, Algorithms and Data Structures.

Technical Skills

Programming: Python, Rust, Java, C#, SQL

Machine Learning: supervised regression and classification, semi-supervised learning, teacher-student self-training, clustering and mixture models, neural networks, feature engineering, walk-forward and cross-validation

Mathematics and Optimization: probability and statistical inference, linear algebra, numerical methods, constrained, combinatorial, and multi-objective optimization, evolutionary, memetic, and swarm algorithms

Infrastructure: Git, Linux, AWS, Databricks, Alpaca API, self-hosted deployment

Projects

ML Equity Selection System 2025–present

- Running live since June 2026, trading multiple accounts through the Alpaca API on a Linux server I built. Scores the full S&P 500 after each close and queues orders for the next open without manual intervention. Model and policy were frozen before deployment.
- Async Rust collection of price history, sector context, SEC filings, news sentiment, and market-regime signals across the S&P 500, cross-sectionally z-scored by date.
- Ridge and logistic regression built in NumPy and blended into a validation-selected score, with the execution policy tuned by grid search on held-out folds.
- Evaluated by 15-fold walk-forward backtesting over 915 out-of-sample trading days and 683 trades, with spread, slippage, and fees modeled, reaching 44.9% CAGR at 1.61 Sharpe and 26.9% maximum drawdown, against 22.2% CAGR at 1.38 Sharpe for SPY.

Home Care Nurse Routing 2026

- Memetic algorithm minimizing nurse travel time under capacity, time-window, and work-hour constraints, combining edge-recombination crossover and adaptive generalized crowding with a parallelized iterated local search phase.
- Mutation rate and selection pressure driven by population entropy to escape premature convergence. Often beat the best known solution, and always within 5% of benchmark across ten seeded runs. Full marks.

Pseudo-Boolean Optimization and Landscape Analysis 2026

- Compared a genetic algorithm, NSGA-II, and binary particle swarm optimization on feature-selection problems maximizing accuracy under a penalty on active features, across search spaces up to 2^{31}. Characterized the landscapes with one-bit Hamming neighborhood analysis and local optima networks. Full marks.

Experience

Aize | Software Engineer Summer Intern, Oslo Jun 2026 – Aug 2026

- Sole ML developer on a data-quality application over large-scale oil and gas asset data. The datasets contained very few usable labels, so this had to be accounted for in the ML architectural choices. I then integrated both models into existing Databricks pipelines.
- Semi-supervised datatype classifier built by teacher-student self-training. A teacher ensemble of a value model and a character and word n-gram text model (both logistic regression) trained on custom-made gold labels, then pseudo-labeled a large pool of unlabeled data. High-confidence predictions trained a student of the same architecture, which beat a fully supervised gold-only baseline.
- Unsupervised expected-attribute predictor. A Bernoulli mixture model clustered assets by property co-occurrence and naive Bayes learned each cluster's metadata signature, so unseen assets could be classified from metadata and their missing properties inferred.